BrainAge Connect

Postman Collection Documentation

A complete local reference generated from the importable Postman collection and observed examples.

Student	Module	Assessment
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PROVENANCE: locally typeset from a Postman Collection v2.1 export, enriched with real Newman examples. It is not a Postman desktop-generated printout or public Postman documentation URL. Native export remains to be obtained in the application for strict compliance with that wording.

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Application: infant MRI metadata and user-supplied precomputed result management. No MRI inference or clinical diagnosis is claimed.

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1. Import and run

Import postman/BrainAgeConnect.postman_collection.json and postman/LocalDemo.postman_environment.json into Postman. Select the local demo environment, start START_DEMO.cmd and run the complete collection in order. Demo administrator: demo.admin / DemoAdmin!2026; demonstration researcher: demo.researcher / DemoResearch!2026. These credentials are only for the synthetic fixture database.

For isolated repeatable evidence, run python -m scripts.run_postman. The installed Newman runner requires Node and its local dependency. Requests obtain tokens and IDs automatically; saved-example IDs are disposable and should not be pasted into live requests.

2. Authentication and resource semantics

Only health, schema, registration and login are public. Approved accounts use a Bearer token. Administrators manage other users. Private samples are invisible to other researchers. Sample writes require owner/admin; nested writes require author/admin. Credentials, auth_version and password hashes are never returned as user resource fields.

JSON requests reject unknown fields and wrong types/ranges. Public IDs are 24-hex strings and dates are UTC ISO 8601. Responses use data/pagination or error.status/error.message. PUT/PATCH are partial updates; successful DELETE/logout returns no body. Child arrays are capped at 200. Statistics respect the same visibility predicate as record reads.

3. Parameters and input schemas

SampleInput

Field	Validation
sample_code	string; maxLength: 120; pattern: ^[A-Za-z0-9_.-]+$
age_at_scan_weeks	number; nullable; minimum: 20; maximum: 80
birth_age_weeks	number; nullable; minimum: 20; maximum: 45
sex	string; allowed: Female, Male, Unknown
has_dmri	boolean
has_fmri	boolean
status	string; allowed: registered, ready, processing, completed, failed
visibility	string; allowed: private, shared
description	string; maxLength: 2000

NoteInput

Field	Validation
text	string; minLength: 1; maxLength: 4000
category	string; allowed: research, quality, review

AnalysisInput

Field	Validation
modality	string; allowed: dmri, fmri, combined
estimated_age_weeks	number; minimum: 20; maximum: 80
method	string; minLength: 1; maxLength: 120
model_version	string; minLength: 1; maxLength: 120
provenance	string; minLength: 1; maxLength: 1000
quality_score	number; nullable; minimum: 0; maximum: 1

Login

Field	Validation
username	string
password	string; format: password

Register

Field	Validation
username	string
password	string; format: password
display_name	string

PasswordChange

Field	Validation
current_password	string; format: password
new_password	string; format: password

UserUpdate

Field	Validation
active	boolean
role	string; allowed: admin, researcher

4. Complete operation reference

POST /api/v1/auth/register

Access: Public.

New accounts are inactive researchers. Reject role escalation; approval is required before login.

JSON body: username, password, display_name. Required: username, password, display_name.

Observed example: 04 Register new researcher; HTTP 201.

Request body

```
{
  "username": "pm.{{run_id}}",
  "password": "PostmanPass!2026",
  "display_name": "Postman Test"
}
```


Response body (token values redacted)


```
{
  "data": {
    "_id": "6ae09f8d1852441fce9d6e4a",
    "active": false,
    "created_at": "2026-10-27T12:56:45.172084Z",
    "display_name": "Postman Test",
    "role": "researcher",
    "username": "pm.17931058037217004"
  },
  "message": "Account created. Administrator approval is required."
}
```


POST /api/v1/auth/login

Access: Public.

The response describes the current service. This endpoint does not expose raw MRI files or perform prediction.

JSON body: username, password. Required: username, password.

Observed example: 03 Administrator login; HTTP 200.

Request body


```
{
  "username": "{{admin_username}}",
  "password": "{{admin_password}}"
}
```


Response body (token values redacted)


```
{
  "access_token": "<redacted-token>",
  "expires_in": 3600,
  "token_type": "Bearer",
  "user": {
    "_id": "6ae09f8887e81a2487ce1416",
    "active": true,
    "created_at": "2026-10-27T12:56:40.711000Z",
    "display_name": "Synthetic Demo Administrator",
    "role": "admin",
    "username": "demo.admin"
  }
}
```


POST /api/v1/auth/logout

Access: Approved user; visible samples only.

Revokes the current jti immediately. The TTL index eventually removes expired revocation records; no response body is returned.

Observed example: 45 Logout; HTTP 204.

Response body (token values redacted)

[No body]

GET /api/v1/auth/me

Access: Approved user; visible samples only.

The response describes the current service. This endpoint does not expose raw MRI files or perform prediction.

Observed example: 08 Current profile; HTTP 200.

Response body (token values redacted)


```
{
  "data": {
    "_id": "6ae09f8d1852441fce9d6e4a",
    "active": true,
    "created_at": "2026-10-27T12:56:45.172000Z",
    "display_name": "Postman Test",
    "role": "researcher",
```

```
    "username": "pm.17931058037217004"
  }
}
```


PUT /api/v1/auth/password

Access: Approved user; visible samples only.

Changing the password increments auth_version, invalidating all existing sessions. Passwords: at least 10 characters, at most 72 UTF-8 bytes.

JSON body: current_password, new_password. Required: current_password, new_password.

Observed example: 42 Change password; HTTP 200.

Request body


```
{
  "current_password": "PostmanPass!2026",
  "new_password": "PostmanChanged!2026"
}
```


Response body (token values redacted)


```
{
  "message": "Password changed. All existing sessions are now invalid."
}
```


GET /api/v1/samples

Access: Approved user; visible samples only.

A session record is the main resource. Creation assigns caller ownership, manual source and private visibility by default. Imported identity and modality-presence fields cannot be changed through the API. Changing reference age recomputes derived differences.

Parameter	Location	Definition
q	query	string
age_min	query	number; minimum: 20; maximum: 80
age_max	query	number; minimum: 20; maximum: 80
modality	query	string; allowed: dmri, fmri, both, neither
source	query	string; allowed: dhcp_local, manual, synthetic
status	query	string; allowed: registered, ready, processing, completed, failed
sex	query	string; allowed: Female, Male, Unknown
page	query	integer; minimum: 1; maximum: 1000000; default: 1
page_size	query	integer; minimum: 1; maximum: 100; default: 20
sort	query	string; allowed: sample_code, age_at_scan_weeks, created_at, updated_at; default: sample_code
order	query	string; allowed: asc, desc; default: asc

Observed example: 14 Search and pagination; HTTP 200.

Response body (token values redacted)


```
{
  "data": [
    {
      "_id": "6ae09f8f1852441fce9d6e4c",
      "age_at_scan_weeks": 40,
      "birth_age_weeks": 38,
      "created_at": "2026-10-27T12:56:47.687000Z",
      "description": "",
      "has_dmri": true,
      "has_fmri": true,
```

```
    "owner_id": "6ae09f8d1852441fce9d6e4a",
    "sample_code": "PM-17931058037217004",
    "sex": "Unknown",
    "source": "manual",
    "status": "registered",
    "updated_at": "2026-10-27T12:56:47.687000Z",
    "version": 1,
    "visibility": "private"
   }
  ],
  "pagination": {
   "page": 1,
   "page_size": 1,
   "pages": 1,
   "total": 1
  }
}
```


POST /api/v1/samples

Access: Approved user; visible samples only.

A session record is the main resource. Creation assigns caller ownership, manual source and private visibility by default. Imported identity and modality-presence fields cannot be changed through the API. Changing reference age recomputes derived differences.

JSON body: sample_code, age_at_scan_weeks, birth_age_weeks, sex, has_dmri, has_fmri, status, visibility, description. Required: sample_code, age_at_scan_weeks.

Observed example: 11 Create private sample; HTTP 201.

Request body


```
{
  "sample_code": "PM-{{run_id}}",
  "age_at_scan_weeks": 40,
  "birth_age_weeks": 38,
  "has_dmri": true,
  "has_fmri": true
}
```


Response body (token values redacted)


```
{
  "data": {
    "_id": "6ae09f8f1852441fce9d6e4c",
    "age_at_scan_weeks": 40,
    "analyses": [],
    "birth_age_weeks": 38,
    "created_at": "2026-10-27T12:56:47.687991Z",
    "description": "",
    "files": [],
    "has_dmri": true,
    "has_fmri": true,
    "notes": [],
    "owner_id": "6ae09f8d1852441fce9d6e4a",
    "sample_code": "PM-17931058037217004",
    "sex": "Unknown",
    "source": "manual",
    "status": "registered",
    "updated_at": "2026-10-27T12:56:47.687991Z",
    "version": 1,
    "visibility": "private"
  }
}
```


GET /api/v1/samples/{sample_id}

Access: Approved user; visible samples only.

A session record is the main resource. Creation assigns caller ownership, manual source and private visibility by default. Imported identity and modality-presence fields cannot be changed through the API. Changing reference age recomputes derived differences.

Parameter	Location	Definition
sample_id	path	string; pattern: ^[a-fA-F0-9]{24}$

Observed example: 13 Retrieve created sample; HTTP 200.

Response body (token values redacted)


```
{
  "data": {
    "_id": "6ae09f8f1852441fce9d6e4c",
    "age_at_scan_weeks": 40,
    "analyses": [],
    "birth_age_weeks": 38,
    "created_at": "2026-10-27T12:56:47.687000Z",
    "description": "",
    "files": [],
    "has_dmri": true,
    "has_fmri": true,
    "notes": [],
    "owner_id": "6ae09f8d1852441fce9d6e4a",
    "sample_code": "PM-17931058037217004",
    "sex": "Unknown",
    "source": "manual",
    "status": "registered",
    "updated_at": "2026-10-27T12:56:47.687000Z",
    "version": 1,
    "visibility": "private"
  }
}
```


PATCH /api/v1/samples/{sample_id}

Access: Owner/admin.

A session record is the main resource. Creation assigns caller ownership, manual source and private visibility by default. Imported identity and modality-presence fields cannot be changed through the API. Changing reference age recomputes derived differences.

Parameter	Location	Definition
sample_id	path	string; pattern: ^[a-fA-F0-9]{24}$

JSON body: sample_code, age_at_scan_weeks, birth_age_weeks, sex, has_dmri, has_fmri, status, visibility, description. Required: one or more editable fields.

No separate observed example for this exact method variant is saved. The documented field/access rules apply; PUT and PATCH share their implemented handler. This statement is not a claim that this variant was exercised in the connected Newman run.

PUT /api/v1/samples/{sample_id}

Access: Owner/admin.

A session record is the main resource. Creation assigns caller ownership, manual source and private visibility by default. Imported identity and modality-presence fields cannot be changed through the API. Changing reference age recomputes derived differences.

Parameter	Location	Definition
sample_id	path	string; pattern: ^[a-fA-F0-9]{24}$

JSON body: sample_code, age_at_scan_weeks, birth_age_weeks, sex, has_dmri, has_fmri, status, visibility, description. Required: one or more editable fields.

Observed example: 18 Partial sample update; HTTP 200.

Request body


```
{
  "description": "Postman lifecycle demonstration"
}
```


Response body (token values redacted)


```
{
  "data": {
    "_id": "6ae09f8f1852441fce9d6e4c",
    "age_at_scan_weeks": 40,
    "analyses": [],
    "birth_age_weeks": 38,
    "created_at": "2026-10-27T12:56:47.687000Z",
    "description": "Postman lifecycle demonstration",
    "files": [],
```

```
    "has_dmri": true,
    "has_fmri": true,
    "notes": [],
    "owner_id": "6ae09f8d1852441fce9d6e4a",
    "sample_code": "PM-17931058037217004",
    "sex": "Unknown",
    "source": "manual",
    "status": "registered",
    "updated_at": "2026-10-27T12:56:48.375000Z",
    "version": 2,
    "visibility": "private"
  }
}
```


DELETE /api/v1/samples/{sample_id}

Access: Owner/admin.

A session record is the main resource. Creation assigns caller ownership, manual source and private visibility by default. Imported identity and modality-presence fields cannot be changed through the API. Changing reference age recomputes derived differences.

Parameter	Location	Definition
sample_id	path	string; pattern: ^[a-fA-F0-9]{24}$

Observed example: 40 Delete sample and embedded records; HTTP 204.

Response body (token values redacted)

[No body]

GET /api/v1/samples/{sample_id}/notes

Access: Approved user; visible samples only.

Notes have an independent ID, author, category and timestamps. A user who can see a sample can add a note; mutation requires the note author or admin. At most 200 notes per session.

Parameter	Location	Definition
sample_id	path	string; pattern: ^[a-fA-F0-9]{24}$
page	query	integer; minimum: 1; maximum: 1000000; default: 1
page_size	query	integer; minimum: 1; maximum: 100; default: 20
sort	query	string; allowed: created_at, updated_at; default: created_at
order	query	string; allowed: asc, desc; default: asc

Observed example: 20 List notes; HTTP 200.

Response body (token values redacted)


```
{
  "data": [
    {
      "_id": "6ae09f901852441fce9d6e50",
      "author_id": "6ae09f8d1852441fce9d6e4a",
      "category": "research",
      "created_at": "2026-10-27T12:56:48.499000Z",
      "text": "Postman note",
      "updated_at": "2026-10-27T12:56:48.499000Z"
    }
  ],
  "pagination": {
    "page": 1,
    "page_size": 1,
    "pages": 1,
    "total": 1
  }
}
```

POST /api/v1/samples/{sample_id}/notes

Access: Approved user; visible samples only.

Notes have an independent ID, author, category and timestamps. A user who can see a sample can add a note; mutation requires the note author or admin. At most 200 notes per session.

Parameter	Location	Definition
sample_id	path	string; pattern: ^[a-fA-F0-9]{24}$

JSON body: text, category. Required: text.

Observed example: 19 Create embedded note; HTTP 201.

Request body

```
{
  "text": "Postman note",
  "category": "research"
}
```

Response body (token values redacted)

```
{
  "data": {
    "_id": "6ae09f901852441fce9d6e50",
    "author_id": "6ae09f8d1852441fce9d6e4a",
    "category": "research",
    "created_at": "2026-10-27T12:56:48.499401Z",
    "text": "Postman note",
    "updated_at": "2026-10-27T12:56:48.499401Z"
  }
}
```

DELETE /api/v1/samples/{sample_id}/notes/{child_id}

Access: Author/admin.

Notes have an independent ID, author, category and timestamps. A user who can see a sample can add a note; mutation requires the note author or admin. At most 200 notes per session.

Parameter	Location	Definition
sample_id	path	string; pattern: ^[a-fA-F0-9]{24}$
child_id	path	string; pattern: ^[a-fA-F0-9]{24}$

Observed example: 39 Delete note; HTTP 204.

Response body (token values redacted)

```
[No body]
```

GET /api/v1/samples/{sample_id}/notes/{child_id}

Access: Approved user; visible samples only.

Notes have an independent ID, author, category and timestamps. A user who can see a sample can add a note; mutation requires the note author or admin. At most 200 notes per session.

Parameter	Location	Definition
sample_id	path	string; pattern: ^[a-fA-F0-9]{24}$
child_id	path	string; pattern: ^[a-fA-F0-9]{24}$

Observed example: 21 Retrieve note; HTTP 200.

Response body (token values redacted)

```
{
  "data": {
    "_id": "6ae09f901852441fce9d6e50",
    "author_id": "6ae09f8d1852441fce9d6e4a",
    "category": "research",
    "created_at": "2026-10-27T12:56:48.499000Z",
```

```
    "text": "Postman note",
    "updated_at": "2026-10-27T12:56:48.499000Z"
  }
}
```


PATCH /api/v1/samples/{sample_id}/notes/{child_id}

Access: Author/admin.

Notes have an independent ID, author, category and timestamps. A user who can see a sample can add a note; mutation requires the note author or admin. At most 200 notes per session.

Parameter	Location	Definition
sample_id	path	string; pattern: ^[a-fA-F0-9]{24}$
child_id	path	string; pattern: ^[a-fA-F0-9]{24}$

JSON body: text, category. Required: one or more editable fields.

No separate observed example for this exact method variant is saved. The documented field/access rules apply; PUT and PATCH share their implemented handler. This statement is not a claim that this variant was exercised in the connected Newman run.

PUT /api/v1/samples/{sample_id}/notes/{child_id}

Access: Author/admin.

Notes have an independent ID, author, category and timestamps. A user who can see a sample can add a note; mutation requires the note author or admin. At most 200 notes per session.

Parameter	Location	Definition
sample_id	path	string; pattern: ^[a-fA-F0-9]{24}$
child_id	path	string; pattern: ^[a-fA-F0-9]{24}$

JSON body: text, category. Required: one or more editable fields.

Observed example: 22 Edit note; HTTP 200.

Request body


```
{
  "text": "Updated test note"
}
```


Response body (token values redacted)


```
{
  "data": {
    "_id": "6ae09f901852441fce9d6e50",
    "author_id": "6ae09f8d1852441fce9d6e4a",
    "category": "research",
    "created_at": "2026-10-27T12:56:48.499000Z",
    "text": "Updated test note",
    "updated_at": "2026-10-27T12:56:48.808848Z"
  }
}
```


GET /api/v1/samples/{sample_id}/analyses

Access: Approved user; visible samples only.

Store a supplied result, never run a model. Required modality availability is checked. Difference = estimated_age_weeks - age_at_scan_weeks, or null when reference age is unknown. At most 200 analyses per session.

Parameter	Location	Definition
sample_id	path	string; pattern: ^[a-fA-F0-9]{24}$
page	query	integer; minimum: 1; maximum: 1000000; default: 1
page_size	query	integer; minimum: 1; maximum: 100; default: 20

Parameter	Location	Definition
sort	query	string; allowed: created_at, updated_at; default: created_at
order	query	string; allowed: asc, desc; default: asc

Observed example: 24 List analysis history; HTTP 200.

Response body (token values redacted)


```
{
  "data": [
    {
      "_id": "6ae09f901852441fce9d6e53",
      "age_difference_weeks": 1.5,
      "author_id": "6ae09f8d1852441fce9d6e4a",
      "created_at": "2026-10-27T12:56:48.918000Z",
      "estimated_age_weeks": 41.5,
      "method": "Postman invented fixture",
      "modality": "dmri",
      "model_version": "test-1",
      "provenance": "Synthetic test request; not an MRI-derived estimate",
      "quality_score": 0.8,
      "result_type": "user_supplied_precomputed",
      "updated_at": "2026-10-27T12:56:48.918000Z"
    }
  ],
  "pagination": {
    "page": 1,
    "page_size": 10,
    "pages": 1,
    "total": 1
  }
}
```


POST /api/v1/samples/{sample_id}/analyses

Access: Approved user; visible samples only.

Store a supplied result, never run a model. Required modality availability is checked. Difference = estimated_age_weeks - age_at_scan_weeks, or null when reference age is unknown. At most 200 analyses per session.

Parameter	Location	Definition
sample_id	path	string; pattern: ^[a-fA-F0-9]{24}$

JSON body: modality, estimated_age_weeks, method, model_version, provenance, quality_score. Required: modality, estimated_age_weeks, method, model_version, provenance.

Observed example: 23 Create precomputed result; HTTP 201.

Request body


```
{
  "modality": "dmri",
  "estimated_age_weeks": 41.5,
  "method": "Postman invented fixture",
  "model_version": "test-1",
  "provenance": "Synthetic test request; not an MRI-derived estimate",
  "quality_score": 0.8
}
```


Response body (token values redacted)


```
{
  "data": {
    "_id": "6ae09f901852441fce9d6e53",
    "age_difference_weeks": 1.5,
    "author_id": "6ae09f8d1852441fce9d6e4a",
    "created_at": "2026-10-27T12:56:48.918637Z",
    "estimated_age_weeks": 41.5,
    "method": "Postman invented fixture",
    "modality": "dmri",
    "model_version": "test-1",
    "provenance": "Synthetic test request; not an MRI-derived estimate",
    "quality_score": 0.8,
    "result_type": "user_supplied_precomputed",
    "updated_at": "2026-10-27T12:56:48.918637Z"
  }
}
```

DELETE /api/v1/samples/{sample_id}/analyses/{child_id}

Access: Author/admin.

Store a supplied result, never run a model. Required modality availability is checked. Difference = estimated_age_weeks - age_at_scan_weeks, or null when reference age is unknown. At most 200 analyses per session.

Parameter	Location	Definition
sample_id	path	string; pattern: ^[a-fA-F0-9]{24}$
child_id	path	string; pattern: ^[a-fA-F0-9]{24}$

Observed example: 38 Delete analysis; HTTP 204.

Response body (token values redacted)

[No body]

GET /api/v1/samples/{sample_id}/analyses/{child_id}

Access: Approved user; visible samples only.

Store a supplied result, never run a model. Required modality availability is checked. Difference = estimated_age_weeks - age_at_scan_weeks, or null when reference age is unknown. At most 200 analyses per session.

Parameter	Location	Definition
sample_id	path	string; pattern: ^[a-fA-F0-9]{24}$
child_id	path	string; pattern: ^[a-fA-F0-9]{24}$

Observed example: 25 Retrieve analysis; HTTP 200.

Response body (token values redacted)


```
{
  "data": {
    "_id": "6ae09f901852441fce9d6e53",
    "age_difference_weeks": 1.5,
    "author_id": "6ae09f8d1852441fce9d6e4a",
    "created_at": "2026-10-27T12:56:48.918000Z",
    "estimated_age_weeks": 41.5,
    "method": "Postman invented fixture",
    "modality": "dmri",
    "model_version": "test-1",
    "provenance": "Synthetic test request; not an MRI-derived estimate",
    "quality_score": 0.8,
    "result_type": "user_supplied_precomputed",
    "updated_at": "2026-10-27T12:56:48.918000Z"
  }
}
```


PATCH /api/v1/samples/{sample_id}/analyses/{child_id}

Access: Author/admin.

Store a supplied result, never run a model. Required modality availability is checked. Difference = estimated_age_weeks - age_at_scan_weeks, or null when reference age is unknown. At most 200 analyses per session.

Parameter	Location	Definition
sample_id	path	string; pattern: ^[a-fA-F0-9]{24}$
child_id	path	string; pattern: ^[a-fA-F0-9]{24}$

JSON body: modality, estimated_age_weeks, method, model_version, provenance, quality_score. Required: one or more editable fields.

No separate observed example for this exact method variant is saved. The documented field/access rules apply; PUT and PATCH share their implemented handler. This statement is not a claim that this variant was exercised in the connected Newman run.

PUT /api/v1/samples/{sample_id}/analyses/{child_id}

Access: Author/admin.

Store a supplied result, never run a model. Required modality availability is checked. Difference = estimated_age_weeks - age_at_scan_weeks, or null when reference age is unknown. At most 200 analyses per session.

Parameter	Location	Definition
sample_id	path	string; pattern: ^[a-fA-F0-9]{24}$
child_id	path	string; pattern: ^[a-fA-F0-9]{24}$

JSON body: modality, estimated_age_weeks, method, model_version, provenance, quality_score. Required: one or more editable fields.

Observed example: 26 Edit analysis; HTTP 200.

Request body


```
{
  "estimated_age_weeks": 42
}
```


Response body (token values redacted)


```
{
  "data": {
    "_id": "6ae09f901852441fce9d6e53",
    "age_difference_weeks": 2,
    "author_id": "6ae09f8d1852441fce9d6e4a",
    "created_at": "2026-10-27T12:56:48.918000Z",
    "estimated_age_weeks": 42,
    "method": "Postman invented fixture",
    "modality": "dmri",
    "model_version": "test-1",
    "provenance": "Synthetic test request; not an MRI-derived estimate",
    "quality_score": 0.8,
    "result_type": "user_supplied_precomputed",
    "updated_at": "2026-10-27T12:56:49.197933Z"
  }
}
```


GET /api/v1/statistics/overview

Access: Approved user; visible samples only.

The response describes the current service. This endpoint does not expose raw MRI files or perform prediction.

Parameter	Location	Definition
q	query	string
age_min	query	number; minimum: 20; maximum: 80
age_max	query	number; minimum: 20; maximum: 80
modality	query	string; allowed: dmri, fmri, both, neither
source	query	string; allowed: dhcp_local, manual, synthetic
status	query	string; allowed: registered, ready, processing, completed, failed
sex	query	string; allowed: Female, Male, Unknown

Observed example: 29 Overview aggregation; HTTP 200.

Response body (token values redacted)


```
{
  "data": {
    "by_source": [
      {
        "_id": "manual",
        "count": 1
      }
    ],
    "by_status": [
      {
        "_id": "registered",
        "count": 1
      }
```

```
    ],
    "modalities": [
      {
        "_id": null,
        "both": 1,
        "dmri": 1,
        "fmri": 1
      }
    ],
    "total": 1
  }
}
```


GET /api/v1/statistics/cohorts

Access: Approved user; visible samples only.

Counts are scan sessions. Age bins: <32, 32-<37, 37-<42, >=42 weeks and unknown; sources separated. Null ages do not contribute to mean scan age. CSV includes a header and Content-Disposition.

Parameter	Location	Definition
q	query	string
age_min	query	number; minimum: 20; maximum: 80
age_max	query	number; minimum: 20; maximum: 80
modality	query	string; allowed: dmri, fmri, both, neither
source	query	string; allowed: dhcp_local, manual, synthetic
status	query	string; allowed: registered, ready, processing, completed, failed
sex	query	string; allowed: Female, Male, Unknown

Observed example: 30 Age cohort aggregation; HTTP 200.

Response body (token values redacted)


```
{
  "data": [
    {
      "_id": {
        "age_group": "37-<42 weeks",
        "source": "manual"
      },
      "dmri_count": 1,
      "fmri_count": 1,
      "mean_scan_age_weeks": 40.0,
      "sample_count": 1
    }
  ],
  "unit": "postmenstrual weeks at scan"
}
```


GET /api/v1/statistics/maturity

Access: Approved user; visible samples only.

Select optional method and version before newest CREATED result per session/modality. Optional age bands and non-null difference counts are available. Unfiltered models may be mixed.

Parameter	Location	Definition
q	query	string
age_min	query	number; minimum: 20; maximum: 80
age_max	query	number; minimum: 20; maximum: 80
modality	query	string; allowed: dmri, fmri, both, neither

Parameter	Location	Definition
source	query	string; allowed: dhcp_local, manual, synthetic
status	query	string; allowed: registered, ready, processing, completed, failed
sex	query	string; allowed: Female, Male, Unknown
method	query	string; maxLength: 120
model_version	query	string; maxLength: 120
group_by	query	string; allowed: overall, age_group; default: overall

Observed example: 31 Maturity aggregation; HTTP 200.

Response body (token values redacted)


```
{
  "data": [
    {
      "_id": {
        "modality": "dmri",
        "source": "manual"
      },
      "mean_age_difference_weeks": 2.0,
      "mean_estimated_age_weeks": 42.0,
      "sample_count": 1
    },
    {
      "_id": {
        "modality": "fmri",
        "source": "manual"
      },
      "mean_age_difference_weeks": 0.5,
      "mean_estimated_age_weeks": 40.5,
      "sample_count": 1
    }
  ],
  "interpretation": "Descriptive summary of supplied results; no diagnostic classification."
}
```


GET /api/v1/statistics/comparison

Access: Approved user; visible samples only.

Paired sessions only: optional separate method/version filters precede newest dMRI and fMRI selection. Grouped by source; unfiltered models may be mixed.

Parameter	Location	Definition
q	query	string
age_min	query	number; minimum: 20; maximum: 80
age_max	query	number; minimum: 20; maximum: 80
modality	query	string; allowed: dmri, fmri, both, neither
source	query	string; allowed: dhcp_local, manual, synthetic
status	query	string; allowed: registered, ready, processing, completed, failed
sex	query	string; allowed: Female, Male, Unknown
dmri_method	query	string; maxLength: 120
dmri_model_version	query	string; maxLength: 120
fmri_method	query	string; maxLength: 120

Parameter	Location	Definition
fmri_model_version	query	string; maxLength: 120

Observed example: 32 Paired modality comparison; HTTP 200.

Response body (token values redacted)


```
{
  "comparison": "Latest dMRI minus latest fMRI estimate for paired samples only.",
  "data": [
    {
      "_id": "manual",
      "mean_dmri_minus_fmri_weeks": 1.5,
      "paired_sample_count": 1
    }
  ]
}
```


GET /api/v1/exports/cohorts

Access: Approved user; visible samples only.

Counts are scan sessions. Age bins: <32, 32-<37, 37-<42, >=42 weeks and unknown; sources separated. Null ages do not contribute to mean scan age. CSV includes a header and Content-Disposition.

Parameter	Location	Definition
q	query	string
age_min	query	number; minimum: 20; maximum: 80
age_max	query	number; minimum: 20; maximum: 80
modality	query	string; allowed: dmri, fmri, both, neither
source	query	string; allowed: dhcp_local, manual, synthetic
status	query	string; allowed: registered, ready, processing, completed, failed
sex	query	string; allowed: Female, Male, Unknown
format	query	string; allowed: json, csv; default: json

Observed example: 33 Export JSON summary; HTTP 200.

Response body (token values redacted)


```
{
  "data": [
    {
      "_id": {
        "age_group": "37-<42 weeks",
        "source": "manual"
      },
      "dmri_count": 1,
      "fmri_count": 1,
      "mean_scan_age_weeks": 40.0,
      "sample_count": 1
    }
  ]
}
```


GET /api/v1/users

Access: Administrator.

Administrator-only accounts. Public JSON excludes password_hash/auth_version. Role/activation changes invalidate sessions. An administrator cannot use this route to alter their own role or activation.

Parameter	Location	Definition
page	query	integer; minimum: 1; maximum: 1000000; default: 1

Parameter	Location	Definition
page_size	query	integer; minimum: 1; maximum: 100; default: 20

Observed example: 35 Administrator users; HTTP 200.

Response body (token values redacted)


```
{
  "data": [
    {
      "_id": "6ae09f8887e81a2487ce1416",
      "active": true,
      "created_at": "2026-10-27T12:56:40.711000Z",
      "display_name": "Synthetic Demo Administrator",
      "role": "admin",
      "username": "demo.admin"
    },
    {
      "_id": "6ae09f8887e81a2487ce1417",
      "active": true,
      "created_at": "2026-10-27T12:56:41.011000Z",
      "display_name": "Synthetic Demo Researcher",
      "role": "researcher",
      "username": "demo.researcher"
    },
    {
      "_id": "6ae09f8d1852441fce9d6e4a",
      "active": true,
      "created_at": "2026-10-27T12:56:45.172000Z",
      "display_name": "Postman Test",
      "role": "researcher",
      "username": "pm.17931058037217004"
    }
  ],
  "pagination": {
    "page": 1,
    "page_size": 100,
    "pages": 1,
    "total": 3
  }
}
```


GET /api/v1/users/{user_id}

Access: Administrator.

Administrator-only accounts. Public JSON excludes password_hash/auth_version. Role/activation changes invalidate sessions. An administrator cannot use this route to alter their own role or activation.

Parameter	Location	Definition
user_id	path	string; pattern: ^[a-fA-F0-9]{24}$

Observed example: 36 Administrator user detail; HTTP 200.

Response body (token values redacted)


```
{
  "data": {
    "_id": "6ae09f8d1852441fce9d6e4a",
    "active": true,
    "created_at": "2026-10-27T12:56:45.172000Z",
    "display_name": "Postman Test",
    "role": "researcher",
    "username": "pm.17931058037217004"
  }
}
```


PATCH /api/v1/users/{user_id}

Access: Administrator.

Administrator-only accounts. Public JSON excludes password_hash/auth_version. Role/activation changes invalidate sessions. An administrator cannot use this route to alter their own role or activation.

Parameter	Location	Definition
user_id	path	string; pattern: ^[a-fA-F0-9]{24}$

JSON body: active, role. Required: one or more editable fields.

Observed example: 47 Deactivate test account; HTTP 200.

Request body

```
{
  "active": false
}
```

Response body (token values redacted)

```
{
  "data": {
    "_id": "6ae09f8d1852441fce9d6e4a",
    "active": false,
    "created_at": "2026-10-27T12:56:45.172000Z",
    "display_name": "Postman Test",
    "role": "researcher",
    "username": "pm.17931058037217004"
  }
}
```

PUT /api/v1/users/{user_id}

Access: Administrator.

Administrator-only accounts. Public JSON excludes password_hash/auth_version. Role/activation changes invalidate sessions. An administrator cannot use this route to alter their own role or activation.

Parameter	Location	Definition
user_id	path	string; pattern: ^[a-fA-F0-9]{24}$

JSON body: active, role. Required: one or more editable fields.

Observed example: 06 Approve researcher; HTTP 200.

Request body

```
{
  "active": true
}
```

Response body (token values redacted)

```
{
  "data": {
    "_id": "6ae09f8d1852441fce9d6e4a",
    "active": true,
    "created_at": "2026-10-27T12:56:45.172000Z",
    "display_name": "Postman Test",
    "role": "researcher",
    "username": "pm.17931058037217004"
  }
}
```

GET /api/v1/audit-logs

Access: Administrator.

Read-only paginated mutation events. Resource writes and audit writes are separate operations on the standalone database, not an atomic cross-collection transaction.

Parameter	Location	Definition
page	query	integer; minimum: 1; maximum: 1000000; default: 1
page_size	query	integer; minimum: 1; maximum: 100; default: 20

Observed example: 37 Administrator audit history; HTTP 200.

Response body (token values redacted)

```
{
  "data": [
```

```
    {
      "_id": "6ae09f911852441fce9d6e57",
      "action": "analyses_created",
      "actor_id": "6ae09f8d1852441fce9d6e4a",
      "created_at": "2026-10-27T12:56:49.401000Z",
      "resource": "samples",
      "resource_id": "6ae09f8f1852441fce9d6e4c"
    },
    {
      "_id": "6ae09f911852441fce9d6e55",
      "action": "analyses_updated",
      "actor_id": "6ae09f8d1852441fce9d6e4a",
      "created_at": "2026-10-27T12:56:49.199000Z",
      "resource": "samples",
      "resource_id": "6ae09f8f1852441fce9d6e4c"
    },
    {
      "_id": "6ae09f901852441fce9d6e54",
      "action": "analyses_created",
      "actor_id": "6ae09f8d1852441fce9d6e4a",
      "created_at": "2026-10-27T12:56:48.919000Z",
      "resource": "samples",
      "resource_id": "6ae09f8f1852441fce9d6e4c"
    },
    {
      "_id": "6ae09f901852441fce9d6e52",
      "action": "notes_updated",
      "actor_id": "6ae09f8d1852441fce9d6e4a",
      "created_at": "2026-10-27T12:56:48.809000Z",
      "resource": "samples",
      "resource_id": "6ae09f8f1852441fce9d6e4c"
    },
    {
      "_id": "6ae09f901852441fce9d6e51",
      "action": "notes_created",
      "actor_id": "6ae09f8d1852441fce9d6e4a",
      "created_at": "2026-10-27T12:56:48.501000Z",
      "resource": "samples",
      "resource_id": "6ae09f8f1852441fce9d6e4c"
    },
    {
      "_id": "6ae09f901852441fce9d6e4f",
      "action": "updated",
      "actor_id": "6ae09f8d1852441fce9d6e4a",
      "created_at": "2026-10-27T12:56:48.377000Z",
      "resource": "samples",
      "resource_id": "6ae09f8f1852441fce9d6e4c"
    },
    {
      "_id": "6ae09f8f1852441fce9d6e4d",
      "action": "created",
      "actor_id": "6ae09f8d1852441fce9d6e4a",
      "created_at": "2026-10-27T12:56:47.689000Z",
      "resource": "samples",
      "resource_id": "6ae09f8f1852441fce9d6e4c"
    },
    {
      "_id": "6ae09f8e1852441fce9d6e4b",
      "action": "permissions_updated",
      "actor_id": "6ae09f8887e81a2487ce1416",
      "created_at": "2026-10-27T12:56:46.282000Z",
      "resource": "users",
      "resource_id": "6ae09f8d1852441fce9d6e4a"
[Excerpt ends here; the complete saved example is in the Postman collection.]
```


GET /api/v1/health

Access: Public.

The response describes the current service. This endpoint does not expose raw MRI files or perform prediction.

Observed example: 01 Health; HTTP 200.

Response body (token values redacted)


```
{
  "application": "BrainAge Connect",
  "status": "ok",
  "version": "1.0.0"
}
```

GET /api/v1/openapi.json

Access: Public.

The response describes the current service. This endpoint does not expose raw MRI files or perform prediction.

Observed example: 48 API schema available; HTTP 200.

Response body (token values redacted)


```
{
  "openapi": "3.0.3",
  "info": {
    "title": "BrainAge Connect",
    "version": "1.0.0",
    "description": "Research metadata and precomputed-result management. No MRI-to-brain-age inference.
All ages are postmenstrual weeks. See API.md for permissions and examples."
  },
  "servers": [
    {
      "url": "http://127.0.0.1:5000"
    },
    {
      "url": "http://127.0.0.1:5001"
    }
  ],
  "security": [
    {
      "BearerAuth": []
    }
  ],
  "paths": {
    "/api/v1/auth/register": {
      "post": {
        "operationId": "auth_register_post",
        "summary": "Register",
        "description": "See docs/API.md for data semantics and authorization.",
        "responses": {
          "201": {
            "description": "Success"
          },
          "400": {
            "description": "Invalid input",
            "content": {
              "application/json": {
                "schema": {
                  "$ref": "#/components/schemas/Error"
                }
              }
            }
          },
          "401": {
            "description": "Missing/expired/revoked session",
            "content": {
              "application/json": {
                "schema": {
                  "$ref": "#/components/schemas/Error"
                }
              }
            }
          },
          "403": {
            "description": "Insufficient permission",
            "content": {
              "application/json": {
                "schema": {
                  "$ref": "#/components/schemas/Error"
                }
              }
            }
          },
          "404": {
            "description": "Not found or not visible",
            "content": {
              "application/json": {
                "schema": {
[Excerpt ends here; the complete saved example is in the Postman collection.]
```

5. Errors and interpretation

{"error": {"status": 400, "message": "age_at_scan_weeks must be a finite number between 20 and 80."}}

400 input/ID validation; 401 missing/invalid/revoked session; 403 permission/inactive login; 404 missing or invisible record; 409 duplicate, version conflict, dependent modality or capped history; 413 request size; 415 JSON type; 503 database availability. Applicable exits depend on the individual handler.

6. Native Postman export handoff

The importable collection includes all 48 connected requests, descriptions, request payloads, automated test scripts and one sanitized observed response for each request. In Postman, import both JSON files, select the environment, open collection documentation and print/save its documentation view to PDF. Run the collection and capture the Runner results as the course requires. No account publication or research-data upload has been performed.

This PDF can serve as a readable reference and preparation material. It must not be relabelled as a native Postman printout. The code listing, endpoint summary and automated test PDF are separately supplied.